MTH2008 Real Analysis: Exercise Sheet 1

This Exercise Sheet covers the material presented in Week 1. By completing the exercises, you should develop an understanding of axioms that define the real number, sets and set operations, and the topology of the real line.

1. In each of the following cases determine the supremum and the infimum of S, justifying your answer.

(i) $S = \left\{ 1 + \dfrac{(-1)^n}{n} \,\middle|\, n \in \mathbb{N} \right\}$

(ii) $S = \{e^n \,|\, n \in \mathbb{N} \cup \{0\}\}$

2. Let X be a set of positive real numbers with supremum $\alpha = \sup X \in \mathbb{R}$. Let

$$Y = \{x^2 + \sqrt{3} \,|\, x \in X\}$$

determine the supremum of Y. Justify your answer.

3. Prove the following statements.

(i) For each $x \in \mathbb{R}$ there exists $n \in \mathbb{N}$ such that $x < n$.

(ii) For all $x, y \in \mathbb{R}$, if $x > 0$, then there exists $n \in \mathbb{N}$ such that $y < nx$.

(iii) For all $x \in \mathbb{R}$, if $x > 0$, then there exists $n \in \mathbb{N}$ with $0 < 1/n < x$.

4. Show that $\inf S \leq \sup S$ for any bounded nonempty set S of real numbers, and give necessary and sufficient conditions for equality.

5. Let S and T be non-empty sets of real numbers and define

$$S + T = \{s + t \mid s \in S \text{ and } t \in T\}.$$

Show that

$$\sup(S + T) = \sup S + \sup T$$

if S and T are bounded above.

6. Let $\mathcal{F}$ be a collection of subsets of $\mathbb{R}$ and define

$$I = \cap\{F \mid F \in \mathcal{F}\} \quad \text{and} \quad U = \cup\{F \mid F \in \mathcal{F}\}.$$

Prove that

(i) $I^c = \cup\{F^c \mid F \in \mathcal{F}\}$, and

(ii) $U^c = \cap\{F^c \mid F \in \mathcal{F}\}$.

These two equalities are sometimes called *De Morgan's law*.

7. In each of the following cases, find the largest ϵ such that S contains an ϵ-neighborhood of x_0.

(i) $x_0 = 3/4 \quad S = \left[\frac{1}{2}, 1\right)$

(ii) $x_0 = 1 \quad S = (0, 2)$

8. Determine if the following sets are open, closed, or neither. Also, determine the set of their interior points, denoted by S^0, the set $(S^c)^0$, and the set $(S^0)^c$.

(i) $S = (-1, 2) \cup [3, \infty)$

(ii) $S = (-\infty, 1) \cup (2, \infty)$

(iii) $S = [-3, -2] \cup [7, 8]$

(iv) $S = \{x \in \mathbb{R} \mid x \in \mathbb{N}\}$

9. (i) Show that the intersection of finitely many open sets is open.

(ii) Give an example showing that the intersection of infinitely many open sets may fail to be open.

10. For each S write down the set of limit points of S and determine the sets ∂S and $\overline{S}$. Moreover, determine the set of isolated points of S as well as the exterior of S.

(i) $S = \mathbb{Z}$

(ii) $S = \bigcup_{n \in \mathbb{Z}} (n, n+1)$

(iii) $S = \{x \mid x = 1/n \text{ for some } n \in \mathbb{N}\}$

11. Let S be a nonempty subset of $\mathbb{R}$ such that if $\mathcal{H}$ is any open covering of S, then S has an open covering $\widetilde{\mathcal{H}}$ comprised of finitely many open sets from $\mathcal{H}$. Show that S is closed and bounded.

12. Let

$$S = \left\{ \frac{(-1)^n}{n} \mid n \in \mathbb{N} \right\},$$

and determine the following, justifying your answer.

(i) The set of interior points of S.

(ii) The set of exterior points of S.

(iii) The set of limit points of S.

(iv) The set of boundary points of S and hence the closure of S.